

Concession Sales Analysis Report

Time Analysis | Market Basket Analysis

Data source: items-2026-01-01-2027-01-01.csv

Executive Summary

Objective

Identify opportunities to increase concession sales revenue by analyzing purchasing time patterns and product associations.

Key Findings

- For hot food items [1], sales peak during the 19:00–19:30 (22.95% of total sales) and 19:30–20:00 (22.78%) sessions, with sales declining after 20:00.
- For popular beer options [2], sales peak during the 19:30–20:00 session and remain relatively strong through 21:30, accounting for 67.83% of total sales, before declining significantly after 21:30.
- A lift of 2.89 indicates that Burgers and Hot Dogs were purchased together about 2.9 times as often as would be expected if the two purchases were independent.
- Hot Dogs and Non-Alcoholic Beverages represented the most frequent pairing, with 6.48% support and a 1.83 lift, indicating a strong opportunity for combo promotions.
- For the customers who purchase Nachos have 52% likelihood of also purchasing a Non-Alcoholic Beverage.

Recommendations

- Focus on increasing hot food inventory and number of available staffs before 19:00 to meet peak demand and reduce resources after 20:00.
- Prioritize beer availability from 19:30–21:30 and ensure sufficient inventory during this period, with reduced replenishment after 21:30 as demand declines.
- Introduce a Hot Dog + Burger combo promotion to capitalize on their strongest association (lift = 2.89) and encourage customers to purchase both items.
- Promote Nachos/Burgers/Hotdogs + Non-Alcoholic Beverage bundles to increase beverage sales through cross-selling.

[1] Cloverdale All-Beef Hot Dog, Cloverdale Brat, Cloverdale Red Dog, M&W Burger, Pulled Pork Sandwich

[2] Bud Light 16 oz Can, Busch Light 16 oz Can Coors Light 16 oz Can, Kona Big Wave Draft, LS Spiked Strawberry Cream Soda, Mango Cart Draft, Michelob Ultra 16 oz Can, Summer Shandy Draft

Business Background

Business Problems

Time Analysis: As the person in charge of the Main Stand concession, which typically has the longest operating hours and the highest customer traffic throughout the game, I noticed that concession demand varies significantly by time period. Analyzing customer traffic across different time sessions through visualization can help game-day staff better plan beer restocking and hot food pre-wrapping. More importantly, understanding demand patterns can provide the kitchen with a better estimate of how many hot food items are needed at the Main Stand during each time period. This can improve preparation efficiency, reduce wait times, and ensure that food is available when demand is highest.

The goal of this project is to transform transaction-level sales data into actionable business recommendations by identifying peak demand periods and improving staffing and inventory planning.

Market Basket Analysis: During the past three months managing the Main Stand concession, I noticed that customers often purchase multiple food and beverage items together, with certain items frequently appearing in the same transaction. Market Basket Analysis can help identify these purchasing patterns and provide opportunities for product bundling and cross-selling. Turning popular item combinations into meal deals can encourage customers to purchase complementary items, increase average transaction value, and generate additional revenue for the Main Stand.

This project is to transform transaction-level sales data into actionable business recommendations by identifying common purchase patterns and opportunities for product bundling and cross-selling.



Dataset Overview

Total transactions: 16755 entries

Products: 41 items

Business Groups:12 groups

Average items/Transaction:1.917

Business Group Mapping

Item	Business Group
Athletic Upside Dawn NA 12 oz Can	Non-Alcoholic Beverage
BD Huckleberry Lemonade	Alcoholic Beverage
BD Strawberry Lemonade	Alcoholic Beverage
Bottled Beverages	Non-Alcoholic Beverage
Bud Light 16 oz Can	Beer
Budweiser 16 oz Can	Beer
Busch Light 16 oz Can	Beer
Candy	Snacks
Carbliss Black Raspberry	Ready-to-Drink Alcohol
Cayman Jack	Ready-to-Drink Alcohol
Chips	Snacks
Cloverdale All-Beef Hot Dog 4:1	Hot Dogs
Cloverdale Brat	Hot Dogs
Cloverdale Red Dog	Hot Dogs
Cookies	Snacks
Coors Light 16 oz Can	Beer
Cotton Candy	Desserts
Cracker Jacks	Snacks
Family Popcorn	Popcorn
Flock Member Popcorn	Popcorn
Goodboy Lemonade (Blueberry)	Ready-to-Drink Alcohol
Ice Cream Sandwich	Desserts
Kona Big Wave Draft	Beer
M&W Burger	Burgers
Mango Cart Draft	Beer
Michelob Ultra 16 oz Can	Beer
Nachos	Nachos
Peanuts	Snacks
Personal Popcorn	Popcorn
Popcorn Helmet	Popcorn
Popsicles	Desserts
Pretzel	Pretzels
Pulled Pork Sandwich	Sandwiches
Small Popcorn	Popcorn
Summer Shandy Draft	Beer
Sun Cruiser	Ready-to-Drink Alcohol
Sun Cruiser Pink Lemonade	Ready-to-Drink Alcohol
Sun Dawgz Spiked Tea	Ready-to-Drink Alcohol
Sunflower Seeds	Snacks
Twisted Tea 12 oz Can	Ready-to-Drink Alcohol
White Claw Black Cherry 12 oz Can	Ready-to-Drink Alcohol

Individual products were consolidated into business groups to reduce data sparsity and improve association analysis.

Sales overview

Food sales summary report

Row Labels	Sum of Net Sales	Count of Item
Brownie	0.01%	0.01%
Candy	3.40%	27.86%
Chips	0.94%	1.33%
Cloverdale All-Beef Hot Dog 4:	11.49%	6.47%
Cloverdale Brat	3.75%	2.24%
Cloverdale Red Dog	6.45%	3.92%
Cloverdale Tangy Snacker	0.01%	0.01%
Coffee	0.02%	0.02%
Cookies	0.58%	0.98%
Cotton Candy	5.57%	2.92%
Cracker Jacks	0.96%	3.38%
Family Popcorn	10.80%	5.37%
Flock Member Popcorn	0.02%	0.04%
M&W Burger	15.65%	6.89%
Nachos	10.56%	4.72%
Peanuts	1.03%	2.51%
Personal Popcorn	0.00%	6.40%
Popcorn Helmet	1.59%	0.57%
Popsicles	1.18%	1.85%
Pretzel	11.96%	7.13%
Pulled Pork Sandwich	2.26%	1.10%
Slushy	1.57%	0.71%
Small Popcorn	8.22%	6.33%
Sunflower Seeds	1.99%	7.24%
Grand Total	100.00%	100.00%

The food items that account for more than 10% of total sales revenue are highlighted in yellow. The data shows that All-Beef Hot Dogs, Family Popcorn, Burgers, Nachos, and Pretzels have the top five sales revenues, collectively accounting for 60.46% of total sales revenue across 24 items.

For sales volume, items that account for more than 5% of total units sold are highlighted in yellow. Candy, All-Beef Hot Dogs, Family Popcorn, Burgers, Pretzels, Small Popcorn, and Sunflower Seeds each represent more than 5% of total sales volume.

Notably, All-Beef Hot Dogs, Family Popcorn, Burgers, and Pretzels are among the top items for both sales' revenue and sales volume. Interestingly, Candy has the highest sales volume, which may be related to the tab policy, as candy is often given away for free with a tab. This could explain why, despite its high sales volume, candy accounts for only 3.40% of total net sales revenue.

Alcohol sales summary report

Row Labels	Sum of Qty
Athletic Upside Dawn NA 12 oz Can	4.28%
Beach Bum	1.72%
Blue Moon 12oz can	0.20%
Bud Light 16 oz Can	6.79%
Bud Light Chelada 16 oz Can	0.03%
Budweiser 16 oz Can	1.82%
Busch Light 16 oz Can	13.79%
Busch Light Apple 12 oz Can	0.07%
Coors Light 16 oz Can	12.60%
Kona Big Wave Draft	11.34%
LS Spiked Strawberry Cream Soda	5.27%
Mango Cart Draft	6.79%
Michelob Ultra 16 oz Can	15.88%
Michelob ULTRA Zero	0.27%
Summer Shandy Draft	12.30%
Sun Dawgz Spiked Tea	0.23%
Twisted Tea 12 oz Can	2.35%
White Claw Black Cherry 12 oz Can	0.27%
Grand Total	100.00%

For the items that account for 5% or more of total sales volume, those meeting the threshold are highlighted in yellow. We can see that Ultra, Busch Light, and Coors Light have the three highest sales volumes, and all three are beer products.

Since both hot food items and popular beer options require frequent restocking, conducting a time-based analysis can provide a better understanding of demand patterns throughout the game. This analysis can help identify peak sales periods and support more efficient inventory and restocking decisions.

Time Analysis

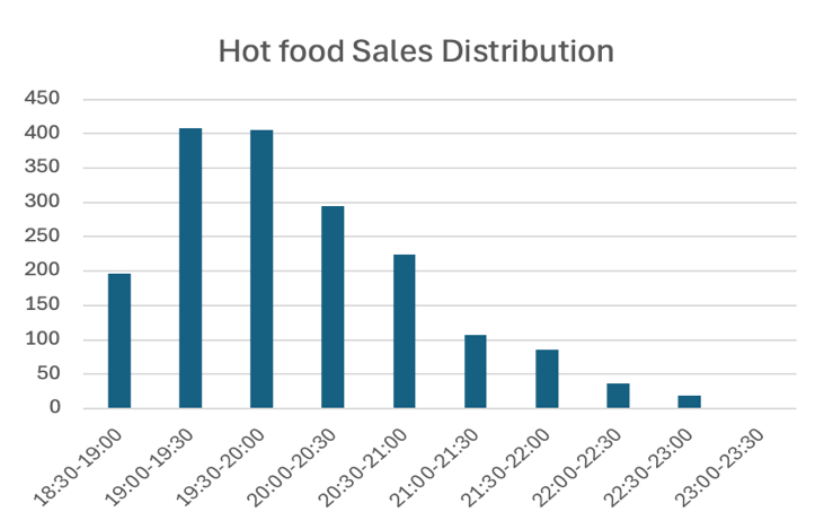
For the time analysis, I created a new column called “Hour Buckets” to group each transaction into a 30-minute time interval. This classification allows us to better understand the number of transactions during each time period and identify demand patterns throughout the game.

Hot food items

Through the pivot table, we can have a clear idea how much hot food transactions happened in each time period.

Row Labels	Percentage	Hour Buckets	Count of the items
18:30-19:00	11.02%	18:30-19:00	196
19:00-19:30	22.95%	19:00-19:30	408
19:30-20:00	22.78%	19:30-20:00	405
20:00-20:30	16.59%	20:00-20:30	295
20:30-21:00	12.60%	20:30-21:00	224
21:00-21:30	6.02%	21:00-21:30	107
21:30-22:00	4.84%	21:30-22:00	86
22:00-22:30	2.08%	22:00-22:30	37
22:30-23:00	1.07%	22:30-23:00	19
23:00-23:30	0.06%	23:00-23:30	1
Grand Total	100.00%		

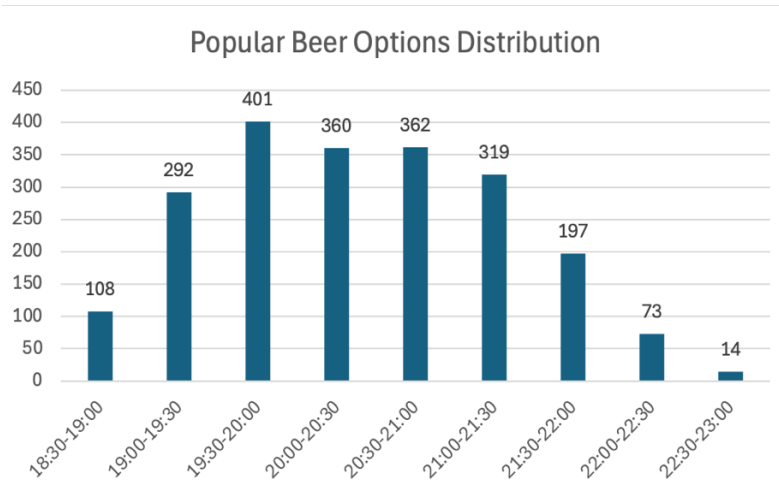
Visualization



Popular Beer

Row Labels	Count of Hour Buckets	Hour Buckets	Count of the items
18:30-19:00	5.08%	18:30-19:00	108
19:00-19:30	13.73%	19:00-19:30	292
19:30-20:00	18.86%	19:30-20:00	401
20:00-20:30	16.93%	20:00-20:30	360
20:30-21:00	17.03%	20:30-21:00	362
21:00-21:30	15.00%	21:00-21:30	319
21:30-22:00	9.27%	21:30-22:00	197
22:00-22:30	3.43%	22:00-22:30	73
22:30-23:00	0.66%	22:30-23:00	14
Grand Total	100.00%		

Visualization:



Market Basket Analysis

We will use the Association Rules Analysis to find the relationships between items that are frequently purchased together in the same transaction, so we can figure out the potential product combinations and cross selling opportunities at the concession stand.

Key Metrics:

Support: Measures how frequent a combination appears in the total transaction

Confidence: Measures how likely customers who purchase A are also likely to purchase B.

Lift: How much more likely A and B got purchased together than what would be expected if they are independent.

Interpretation of lift:

If $\text{lift} > 1$: There's a positive association, these two items are purchased more often than expected

If $\text{lift} = 0$: There's no meaningful association between these two items

If $\text{lift} < 1$: There's a negative association, items are purchased together less often than expected.

Business Takeaway: Lift is a critical metric for identifying which two items have strong associations and therefore present promising opportunities for bundling and cross-selling.

Business Insight:

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift
0	(Hot Dogs)	(Burgers)	0.113959	0.063043	0.020824	0.182731	2.898491
1	(Burgers)	(Hot Dogs)	0.063043	0.113959	0.020824	0.330309	2.898491
2	(Non-Alcoholic Beverage)	(Burgers)	0.310641	0.063043	0.035812	0.115285	1.828666
3	(Burgers)	(Non-Alcoholic Beverage)	0.063043	0.310641	0.035812	0.568058	1.828666
4	(Hot Dogs)	(Non-Alcoholic Beverage)	0.113959	0.310641	0.064760	0.568273	1.829358
5	(Non-Alcoholic Beverage)	(Hot Dogs)	0.310641	0.113959	0.064760	0.208471	1.829358
6	(Nachos)	(Non-Alcoholic Beverage)	0.047140	0.310641	0.024828	0.526699	1.695525
7	(Non-Alcoholic Beverage)	(Nachos)	0.310641	0.047140	0.024828	0.079926	1.695525

(The pairs above all have a lift above 1.5)

Finding 1:

Customers who purchase Burgers, Hot Dogs, or Nachos have a 50%+ likelihood of also purchasing a Non-Alcoholic Beverage.

Business implication: Introduce combo meals pairing Burgers, Hot Dogs, or Nachos with a Non-Alcoholic Beverage to encourage cross-selling and increase average basket size.

Finding 2:

Customers buy Burgers and Hot Dogs together more often than expected, with a lift of 2.89, indicating a strong positive association between the two items.

Business implication: Consider offering a Burger + Hot Dog combo to encourage customers to purchase both items in the same transaction.

Limitations & Future Work

Limitations:

- **Limited Data:** Only half-season sales data is available.
- **Product Granularity:** Market Basket Analysis was conducted at the Business Group level, so the analysis identifies beverage pairing opportunities but not specific soda preferences.
- **No Pricing Experiment:** No data available to assess the impact of combo pricing.

Future Works:

- **Complete Dataset:** Analyze full-season sales data when available.
- **Product-Level Analysis:** Identify the most popular soda within Non-Alcoholic Beverages to determine the best specific drink for the combo.
- **Post-Implementation Analysis:** Evaluate combo performance after implementation.